Class 08: Breast Cancer Mini Project

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Background

In today's class we will be employing all the R techniques for data analysis that we have learned thus far - including the machine learning methods of clustering and PCA - to analyze real breast cancer biopsy data.

Data Input

The data is in the csv format

```
fna.data <- "WisconsinCancer.csv"

wisc.df <- read.csv(fna.data, row.names=1)
```

Q1. How many observations are in this dataset?

```r
nrow(wisc.df)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```r
length(wisc.df$diagnosis == "M")
```

```
[1] 569
```

Q3. How many variables/features in the data are suffixed with _mean?

```r
length(grep("_mean", colnames(wisc.df)))
```

```
[1] 10
```

We need to remove the `diagnosis` column before we do any further analysis of this dataset - we don't want to pass this to PCA etc. We will save it as a seperate wee vector that we can use later to compare our findings to those of experts.

```r
wisc.data <- wisc.df[,-1]
diagnosis <- wisc.df$diagnosis
```

Principal Component Analysis (PCA)

The main function in base R is called `prcomp()` we will use the optional argument `scale=TRUE` here as the data columns /features/dimensionsare on very different scales in the original data set.

```r
wisc.pr <- prcomp(wisc.data, scale = TRUE)
```

```r
attributes(wisc.pr)
```

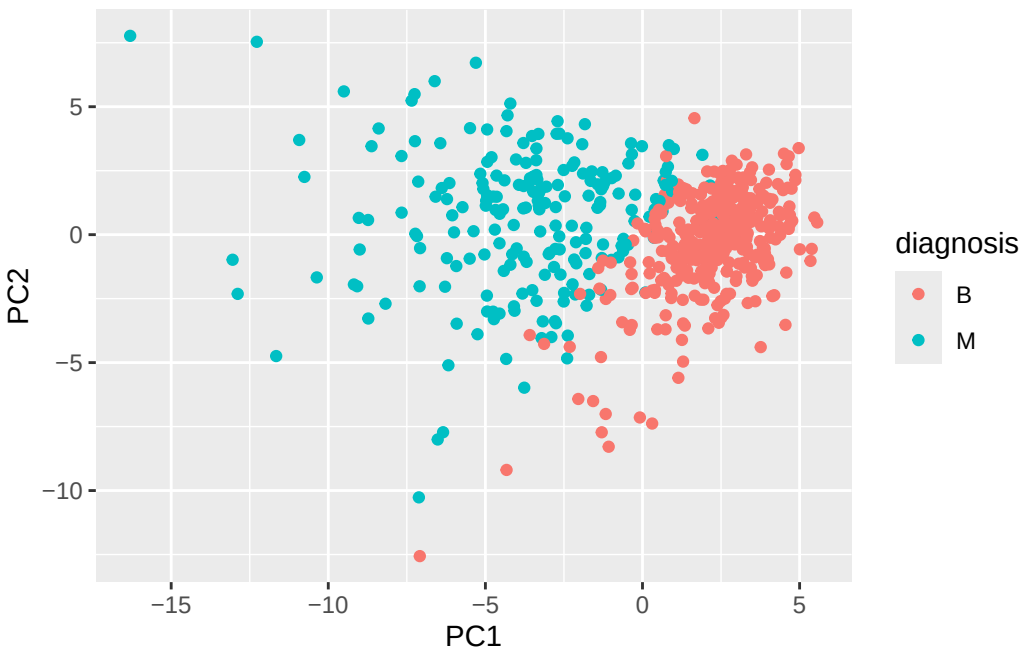
```
$names
[1] "sdev"     "rotation" "center"   "scale"    "x"

$class
[1] "prcomp"
```

Scree Plot

```
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1,PC2, col = diagnosis) +
  geom_point()
```




```
colMeans(wisc.pr$x)
```

```
          PC1           PC2           PC3           PC4           PC5
 2.811264e-15 -8.489596e-16  4.167727e-16  3.523056e-15 -1.179295e-15
          PC6           PC7           PC8           PC9          PC10
-1.627677e-15  6.438903e-16  3.664321e-15  6.270126e-16 -2.888519e-15
         PC11          PC12          PC13          PC14          PC15
-2.093619e-16  1.174466e-15  2.478002e-17  1.195295e-15 -1.966792e-15
         PC16          PC17          PC18          PC19          PC20
-8.893491e-16  1.959085e-15 -4.193482e-15 -3.376229e-15  1.759674e-15
         PC21          PC22          PC23          PC24          PC25
-6.939967e-15  1.980353e-15  1.715943e-16  1.970256e-15  1.372785e-15
         PC26          PC27          PC28          PC29          PC30
 2.178496e-15  1.375828e-15  1.287110e-16  1.171551e-15 -1.512990e-15
```

```r
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

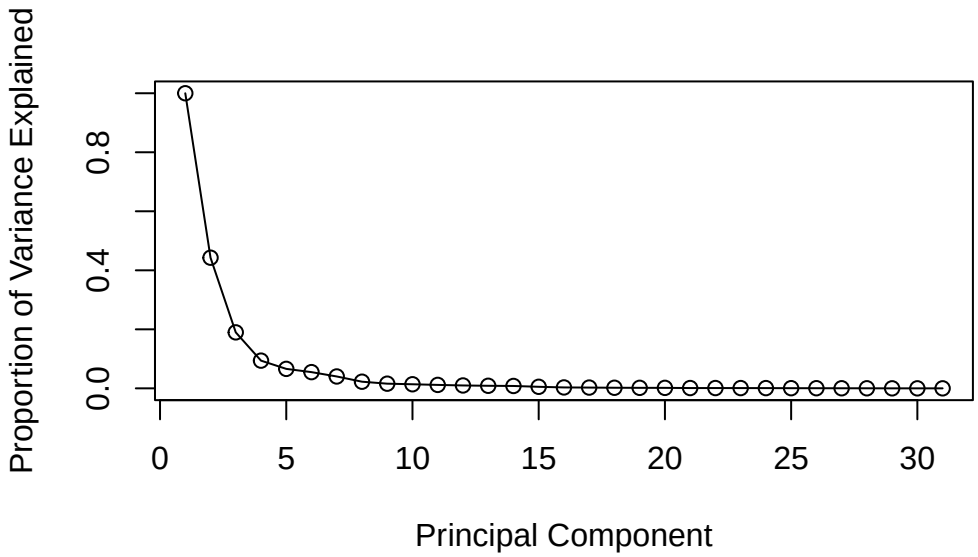
```r
pv.var <- wisc.pr$sdev^2
head(pv.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

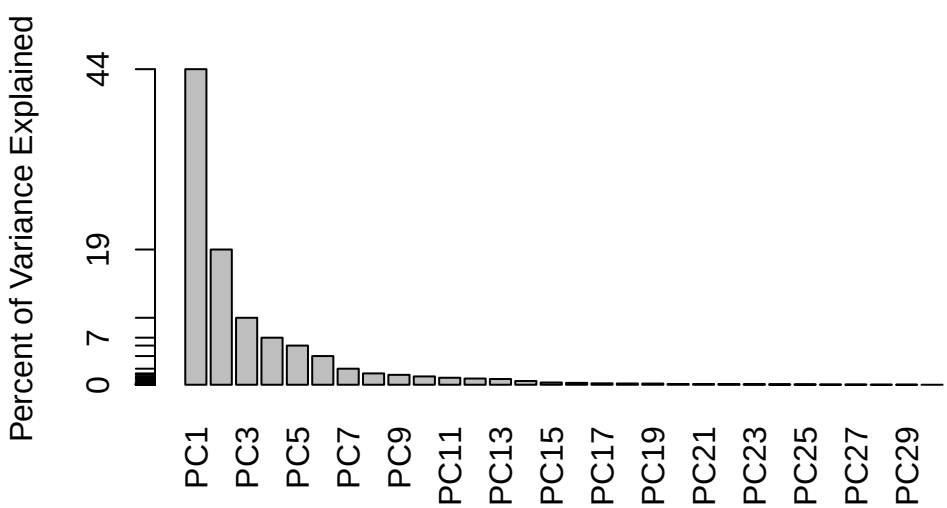
```r
pve <- pv.var/sum(pv.var)
head(pve)
```

```
[1] 0.44272026 0.18971182 0.09393163 0.06602135 0.05495768 0.04024522
```

```r
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```




```
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

0.4427

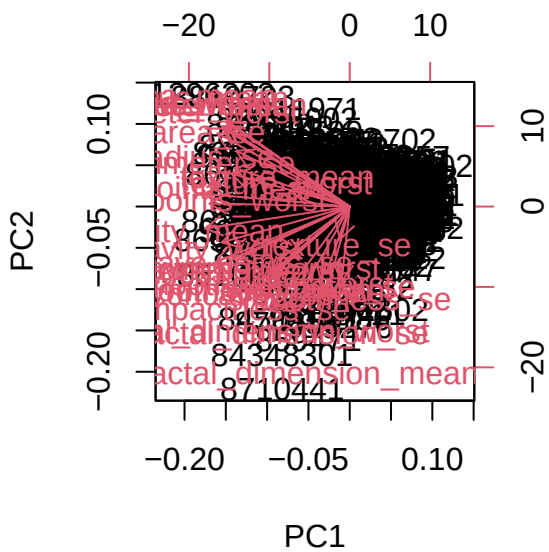
> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

3 > Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7

Interpreting PCA results

```
biplot(wisc.pr)
```



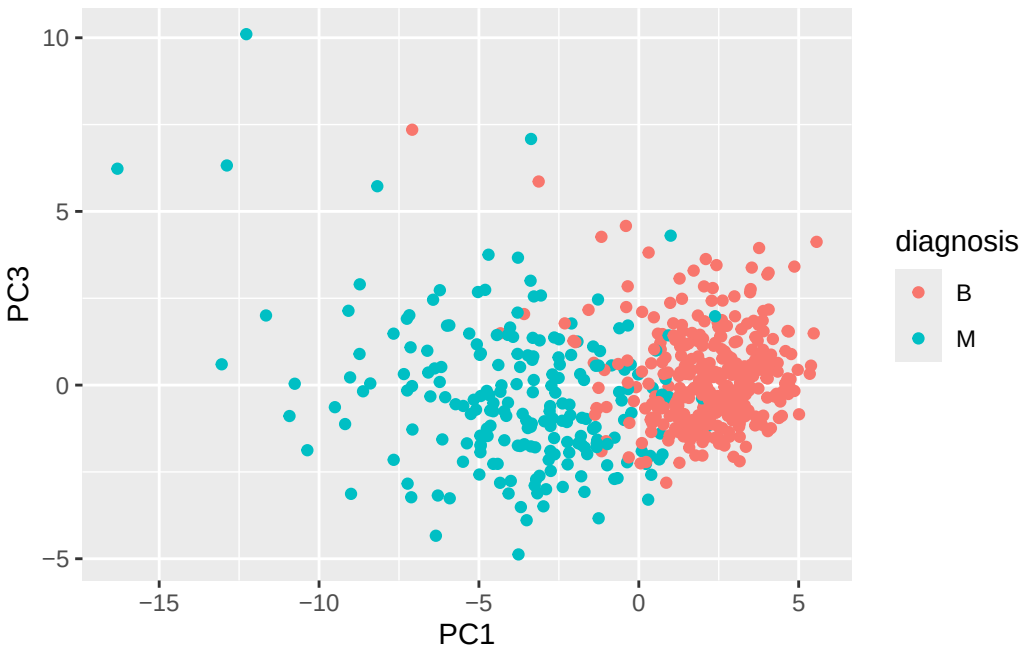
> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

This plot is extremely hard to interpret, it's messy.

Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

It's showing that principal component 1 is accounting for a large portion of the variance, thus capturing the distinct clusters. PC1 captures ~ 44% while PC2 and PC3 are much lower percentages.

```
ggplot(wisc.pr$x) +
  aes(PC1,PC3, col = diagnosis) +
  geom_point()
```



Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

-0.26085376, no, there are no features with larger contributions than this one.

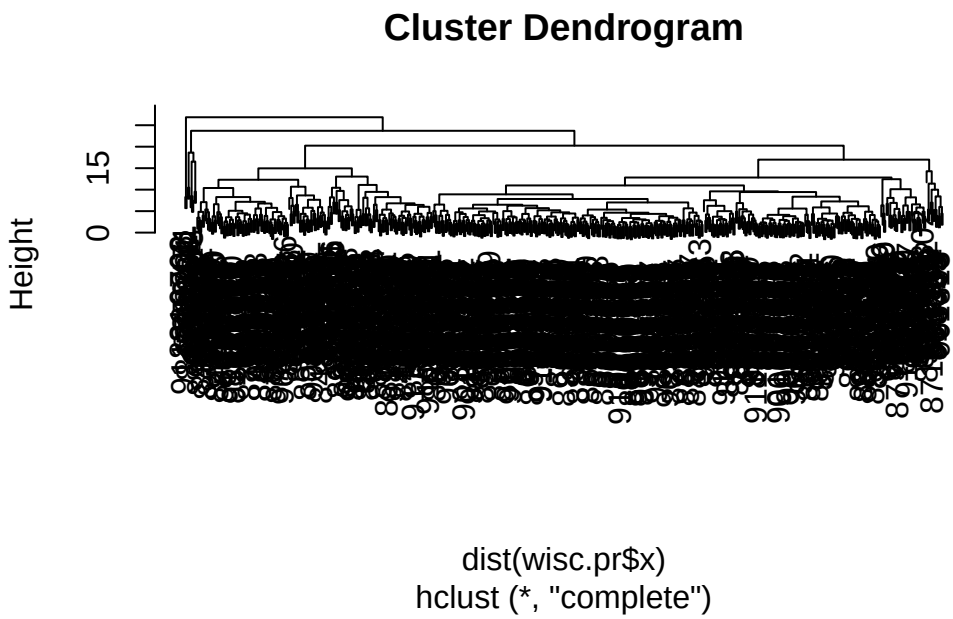
```
wisc.pr$rotation[,1]
```

```
       radius_mean      texture_mean    perimeter_mean
       -0.21890244       -0.10372458       -0.22753729
         area_mean    smoothness_mean  compactness_mean
```

```
          -0.22099499            -0.14258969            -0.23928535
        concavity_mean     concave.points_mean           symmetry_mean
          -0.25840048            -0.26085376            -0.13816696
 fractal_dimension_mean              radius_se             texture_se
          -0.06436335            -0.20597878            -0.01742803
         perimeter_se                 area_se           smoothness_se
          -0.21132592            -0.20286964            -0.01453145
        compactness_se           concavity_se       concave.points_se
          -0.17039345            -0.15358979            -0.18341740
          symmetry_se    fractal_dimension_se            radius_worst
          -0.04249842            -0.10256832            -0.22799663
         texture_worst         perimeter_worst             area_worst
          -0.10446933            -0.23663968            -0.22487053
      smoothness_worst       compactness_worst         concavity_worst
          -0.12795256            -0.21009588            -0.22876753
  concave.points_worst          symmetry_worst fractal_dimension_worst
          -0.25088597            -0.12290456            -0.13178394
```

4. Hierarchical Clustering

```r
data.dist <- hclust(dist(wisc.pr$x))
plot(data.dist)
```

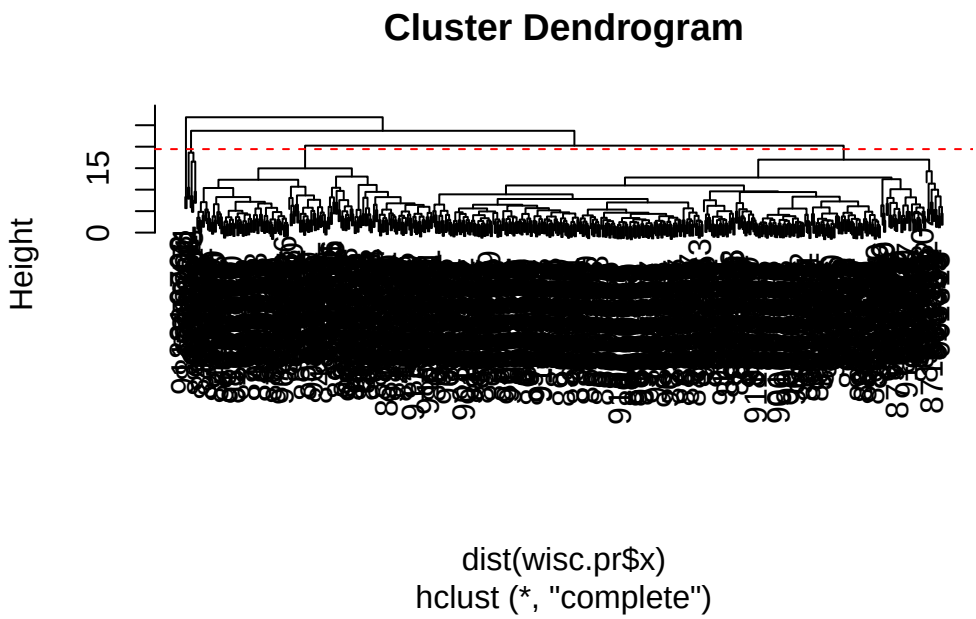


Results

```
plot(data.dist)


n <- length(data.dist$height)
h4 <- mean(data.dist$height[(n-3):(n-2)])

abline(h = h4, col="red", lty=2)
```




```
h4
```

```
[1] 19.43994
```

```
n
```

```
[1] 568
```

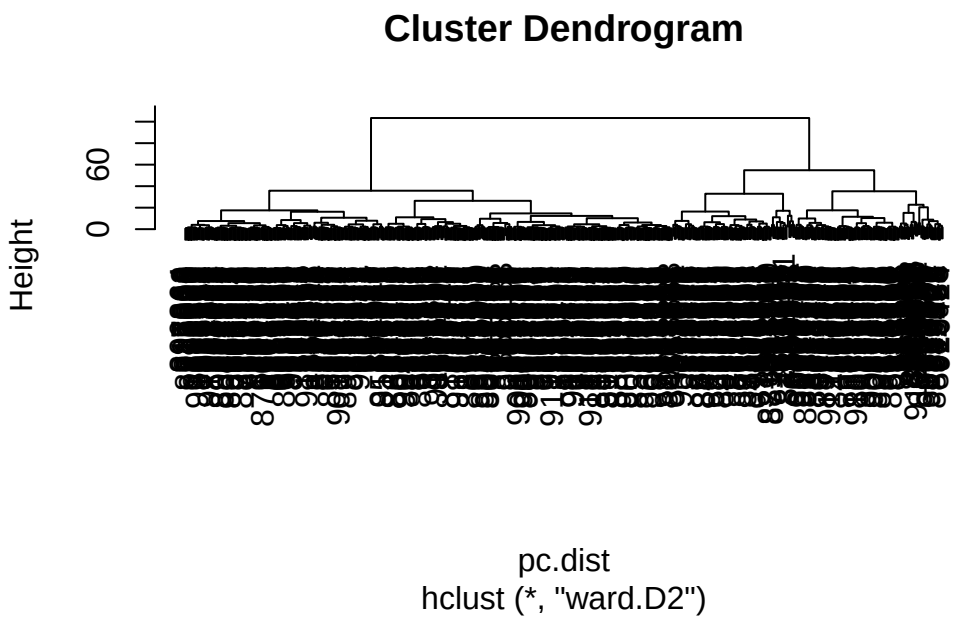
> Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters? 19.43994
>
> Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

`ward.d2` - the variance is minimized

Combining Methods

the idea here is that i can take my ne variables (the PCs) that are better descriptors of the data-set than the original features (i.e. the 30 columns in`wisc.data`) and use these as a basis for clustering.

```r
pc.dist <- dist(wisc.pr$x[ ,1:3])
wisc.pr.hclust <- hclust(pc.dist, method = "ward.D2")
plot(wisc.pr.hclust)
```



Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

```r
grps <- cutree(wisc.pr.hclust, k =2)
table(grps)
```

```
grps
  1   2
203 366
```

I can now run `table()` with both my clustering `grps` and the expert `diagnosis`.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```r
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  24 179
   2 333  33
```

Our cluster "1" has 179 "M" diagnosis Our cluster "2" has 333 "B" diagnosis

179 TP 24 FP 333 TN 33 FN

Sensitivity: TP/(TP+FN)

```r
179/(179+33)
```

```
[1] 0.8443396
```

Specificity TN/(TN+FP)

```r
333/(333+24)
```

```
[1] 0.9327731
```

Predictions

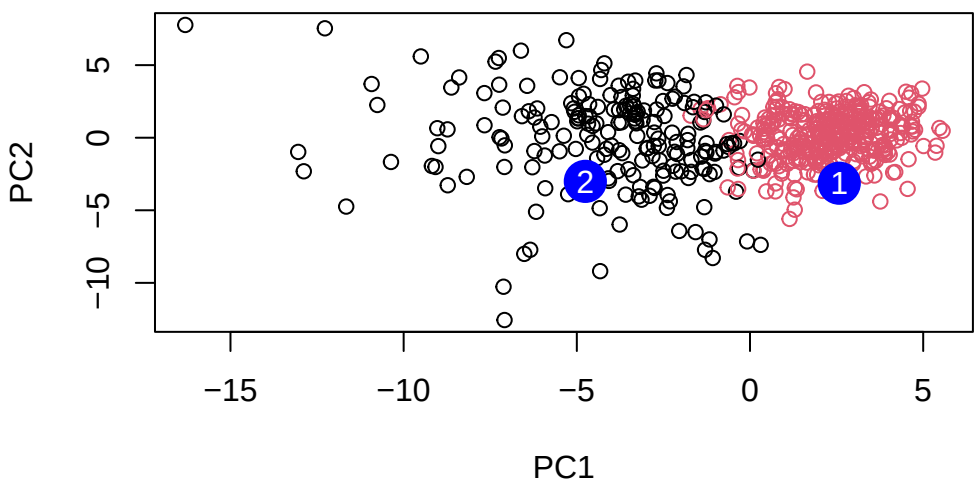
We will use the `predict()` function that will take our PCA model from before and new cancer cell data and project that data onto our PCA space.

```r
new <- read.csv("new_samples.csv")
npc <- predict(wisc.pr, newdata = new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
```

```
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
              PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col = "blue", pch = 16, cex = 3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q.16 Which of these new patients should we prioritize for follow up based on your results?

Patient 2 lines up with the malignent data points. Its far enough awy from the benign cluster that we can also assume that it's most likely not a false positive.